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RegionDiffusion: Generative data augmentation for object detection with diffusion models

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HIGHLIGHTS

- We propose RegionDiffusion, a novel layout-to-image model with outstanding generation and zero-shot capabilities enabled by region self-attention.
- We introduce a comprehensive generative data augmentation pipeline utilizing a series of preprocessing and postprocessing steps, which significantly enhance the performance of object detectors across general and specific domains.

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ABSTRACT

Recently, research on generative data augmentation based on layout-to-image diffusion models for object detection has made some progress. However, there remain challenges in generated data, such as low visual quality, weak generalization and mismatches between generated images and annotations, which hinder the performance of object detectors. To alleviate these issues, we propose the RegionDiffusion, a novel layout-to-image model that uses an attention block called Region Self-Attention to divide the self-attention operation among visual tokens into several independent regions based on the bounding box of the layout to generate objects accurately in the given locations. Then, we design a pipeline utilizing a series of preprocessing and postprocessing steps to improve the quality of the generated data and then train detectors. With extensive experiments, we demonstrate that RegionDiffusion surpasses state-of-the-art models in layout-to-image generation and our pipeline significantly improves the performance of object detectors in both general domains and specific domains.

1. Introduction

In the realm of computer vision, object detection is a fundamental task with wide-ranging applications. The performance of object detection models highly depends on the quality and quantity of training data, but obtaining manually annotated images is both expensive and time-consuming. As an alternative technique, generative data augmentation aiming to expand the dataset and improve model generalization through generative models has great potential to alleviate this problem.

Traditional data augmentation methods for object detection like geometric transformations (including rotation, translation, scaling, and flipping) and image mixing (such as CutMix [1] and MixUp [2]) can only introduce a restricted range of variations to the existing data and lack the ability to generate entirely new and diverse data samples or change the relationships between objects in a natural-looking way. Generative data augmentation refers to leveraging generative models such as GANs [3] or diffusion models [4,5] to generate new and unique data samples. It exposes the object detection model to a wider range of object appearances, poses, and backgrounds.

The process of layout-to-image models [6–8] generating images by leveraging the position coordinates and category descriptions of each object is essentially the inverse of object detection. Early GAN-based layout-to-image models [6,7] suffer from poor image quality and layout accuracy. However, recent diffusion-based models, such as ReCo [9], GLIGEN [8], HiCo [10] and GeoDiffusion [11], have indeed achieved significant improvements compared to early models—they can generate more realistic images and have better layout alignment to some extent. Despite these advancements, they still have critical limitations: (1) They lack effective encoding of layout information, leading to fusion of overlapping instances of the same category; (2) Their attention mechanisms fail to partition regions based on layout, resulting in inaccurate object

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positioning; (3) They tend to over-focus on large instances and background during training, ignoring small-object generation; (4) They often generate extra instances in the background, affecting layout cleanliness.

To tackle these challenges, we propose RegionDiffusion, a novel and powerful layout-to-image model integrating three core components. The Layout Tokenizer encodes each instance's bounding box and category into layout tokens with trainable positional encoding to avoid fusion of overlapping same-category instances and introduces a background token for clean background generation. The Region Self-Attention divides visual tokens into distinct regions based on layout bounding boxes, enabling accurate object generation at specified locations. Area-based Loss Re-weighting uses an area-based mask to adjust loss weights, preventing over-focus on large instances and improving small-object generation.

Existing generative data augmentation methods for object detection, such as those based on diffusion models [11–13], have made progress in generating image-annotation pairs, but they still face critical limitations in practical deployment: (1) Generated data often suffers from instance fusion (e.g., overlapping objects) or extra generated objects not in annotations due to unfiltered complex layouts from real datasets; (2) Lack of domain-specific adjustments, making it difficult to adapt to scenarios like artistic styles (e.g., Clipart [14]); (3) Generated instances may have category mismatches or missing details, which degrade detector training efficiency.

To address these issues, our proposed generative data augmentation pipeline introduces three key innovations tailored for object detection: First, a Front Filter based on NMS is designed to remove highly overlapping bounding boxes from real annotations. This reduces layout complexity and avoids instance fusion during generation—a common issue in crowded scenes that existing methods fail to resolve. Second, a domain-aware prompt construction strategy integrates object categories, scene descriptions, and dataset captions. This ensures generated data matches both layout constraints and domain characteristics, which are not considered in generic pipelines. Third, a CLIP-based Post Filter filters out low-quality instances by measuring similarity between generated objects and their categories. This eliminates incorrect annotations that would otherwise mislead detector training.

From an engineering perspective, this pipeline is modular and easy to deploy-each component (filters, prompt templates) can be adjusted for specific datasets. It also significantly reduces reliance on manual annotations in scenarios where labeled data are scarce. Moreover, it can be easily extended to many downstream tasks such as cross-domain object detection.

Our experimental results show that RegionDiffusion outperforms state-of-the-art models in layout-to-image generation and zero-shot capabilities. The proposed pipeline significantly boosts the performance of YOLOv8 [15] by 6.1 % AP for COCO [16] and up to 18.2 % AP for downstream datasets. Ablation studies verify the effectiveness of each component in RegionDiffusion and the pipeline.

Contributions:

- We propose RegionDiffusion, a novel layout-to-image model with outstanding generation and zero-shot capabilities.
- We introduce a comprehensive generative data augmentation pipeline, which significantly enhances the performance of object detectors in general and specific domains.
- We conduct extensive experiments to validate the superiority of RegionDiffusion and the proposed pipeline. Ablation studies further verify the effectiveness of each component within RegionDiffusion and the pipeline.

2. Related works

2.1. Diffusion models

Diffusion Models [4,5,17] have attracted high-profile notice in generative artificial intelligence in recent years because of their powerful generative capabilities. Through the forward diffusion and reverse denoising processes, they can generate high-quality images and have replaced Variational AutoEncoders (VAEs) [18] and Generative Adversarial Networks (GANs) [3] to become the mainstream method for image generation. Latent diffusion models (LDMs) [19] transform the diffusion process from the pixel space to the latent space, greatly improving the computational efficiency and introducing conditioning inputs such as text descriptions. Large-scale pre-training works [19–24] use web-scale image-text pair data to endow the diffusion model with a powerful ability to generate various realistic images according to text prompts. In addition, [25–27] also introduce diffusion models into other modality fields and have demonstrated powerful capabilities for text, audio and video generation.

2.2. Object detection

Object detection, a core computer vision task, has advanced significantly through diverse methodological innovations. Two-stage frameworks, exemplified by R-CNN [28] and its variants (Fast R-CNN [29], Faster R-CNN [30]), established the paradigm of region proposal generation followed by refinement, achieving high accuracy but with higher computational costs. One-stage detectors like YOLO [31] and SSD [32] streamlined the process into end-to-end regression, enabling real-time inference via multi-scale feature fusion and simplified pipelines. Recent years have seen transformer-based methods such as DETR [33] and Deformable DETR [34] replace hand-designed components with attention mechanisms, enhancing pipeline simplicity and performance. Complementary progress in data augmentation (e.g., Mosaic [35]) and lightweight architectures has further expanded detection applicability across scenarios such as remote sensing detection [36].

2.3. Layout-to-image models

While text-to-image models can condition the generated images using textual prompts, they lack the ability to impose precise spatial control over the generated content, such as accurately specifying the size and position of objects in an image. In contrast, layout-to-image models utilize additional control inputs, such as bounding boxes, masks, points, or scribbles, to guide the spatial layout of generated images. Early works such as [6,7] attempt to implement the layout-to-image models based on GANs. However, due to the limitations of the GANs' generation capability, the quality of the images and the conformity of the layouts are rather restricted, and the control conditions are relatively simple. In recent years, works based on pre-trained diffusion models like ReCo [9], GLIGEN [8], ControlNet [37], HiCo [10], InstanceDiffusion [38] and MIGC [39,40] generate more realistic images with a higher degree of layout conformity. Some of them support adding more complex layout inputs such as points, scribbles, person keypoints and sketches. Works like BoxDiff [41], R&B [42], and GrounDiT [43] endow diffusion models with layout control capabilities without requiring additional training.

2.4. Generative data augmentation

Generative data augmentation relies on generative models to create new sample data, thereby increasing the scale and diversity of the dataset. Recently, numerous works have attempted to utilize diffusion models for data augmentation in visual tasks. [44–47] demonstrate that the images generated by diffusion models can be used to improve the performance of image classification tasks. [48–51] adopt diffusion models with masks as layout conditions to conduct generative data augmentation for segmentation. [52–54] try to generate photo-realistic street-view images using diffusion models to boost the performance of autonomous driving tasks. For the human-object interaction detection task, DIFFUSIONHOI [55] utilizes diffusion models to generate images and pseudo-annotations of specific human-object interactions to enrich training samples, especially for long-tailed and unseen interaction categories. For object detection tasks, [56] generates the required objects and background separately using diffusion models and then pastes objects onto background images. However, the data generated by this method is not natural and harmonious. [11–13] directly add layout conditions to the text prompts to generate images that match the given annotations and use them to train detectors. Nevertheless, since the text encoder and U-Net are fine-tuned, only a limited number of object categories can be generated, and their zero-shot ability is rather poor. [57] employs ControlNet and uses HED [58] visual priors to control the layout and adopts a post-filtering method. However, this strong layout control based on object edges leads to a reduction in the diversity of the generated objects.

3. RegionDiffusion

In this section, we present RegionDiffusion, a novel layout-to-image generation model composed of three main components: Layout Tokenizer, Region Self-Attention and Area-based Loss Re-weighting. As shown in Fig. 1, we first use the Layout Tokenizer to encode layout information into tokens. Then, Region Self-Attention is applied to integrate the layout information into the diffusion model backbone. Finally, we use the Area-based Loss to re-weight the original training loss function during training.

3.1. Layout tokenizer

For every instance I_i, its layout can be represented as a 2-tuple $(\mathbf{b}_i, c_i)$ where $\mathbf{b}_i$ denotes the bounding box in the format $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$, and c_i represents the category of the instance I_i. To integrate layout information into the diffusion model, we use the Layout Tokenizer to convert $(\mathbf{b}_i, c_i)$ into a single layout token $\mathbf{L}_i$. As shown in Fig. 1(b), we employ CLIP [59] to encode the category c_i and use the Fourier embedder to project the bounding box $\mathbf{b}_i$. The resulting text features and positional embeddings are concatenated and fused using a multi-layer perceptron (MLP) to obtain $\mathbf{L}_i$, following the similar approach to [8,38]. Additionally, since the text features obtained via the CLIP encoder are identical for instances of the same category, we add a trainable positional encoding $\mathbf{p}_i$ to the text features to prevent fusing multiple overlapping instances with same category into a single one:

$$\mathbf{L}_i = \mathrm{MLP}([(\mathrm{CLIP}(c_i) + \mathbf{p}_i), \mathrm{Fourier}(\mathbf{b}_i)]), \tag{1}$$

where $[\cdot, \cdot]$ denotes concatenation. Even though layout tokens enable the generation of instances at specified positions, they cannot prevent the generation of new instances in background region. This mismatch results in the generated image containing more instances than specified in the given layout. To address this issue and ensure the model generates a clean background, we introduce a learnable background token $\mathbf{L}_{bg}$, which is initialized as "*null*". This token is used to interact with the background visual tokens that do not belong to any instance in the Region Self-Attention mechanism described below.

3.2. Region self-attention

To enable the original text-to-image diffusion model to have the ability of layout control, [8] uses vanilla self-attention as the fusion mechanism between layout and visual tokens. This approach, where all tokens including visual and layout tokens can globally interact, often causes attribute leakage (the attribute of one instance transfers to another) and spatial leakage (one instance affects the generation position of another) among instances. To address these issues, we introduce Region Self-Attention to interact the layout tokens generated above with the original visual tokens and insert Region Self-Attention after the self-attention module within the U-Net layers:

$$\mathbf{V} = \mathbf{V} + \mathrm{SelfAttn}(\mathbf{V}) \tag{2}$$

$$\mathbf{V} = \mathbf{V} + \text{Region-SelfAttn}([\mathbf{V}, \mathbf{L}]) \tag{3}$$

$$\mathbf{V} = \mathbf{V} + \mathrm{CrossAttn}(\mathbf{V}, \mathbf{G}), \tag{4}$$

where $\mathbf{V}$ denotes the latent visual feature tokens encoded by the VAE, and $\mathbf{L}$ is the layout tokens from the layout tokenizer and $\mathbf{G}$ is the text feature tokens encoded by CLIP of global prompt. As shown in Fig. 1(c), for a U-Net layer with the resolution of $h \times w$, given n instances to be generated and their layout information. (1) We use the bounding box of each instance to partition the $h \times w$ visual tokens $\mathbf{V}$ into n instance regions and a background region, where $\mathbf{V}_i$ represents visual tokens in instance region i, $\mathbf{V}_{bg}$ is visual tokens in background region. (2) For instance region i, self-attention is performed only over the concatenation of visual tokens within the region and the corresponding layout token $[\mathbf{V}_i, \mathbf{L}_i]$. For background region, self-attention is performed only over the concatenation $[\mathbf{V}_{bg}, \mathbf{L}_{bg}]$. (3) Finally, we aggregate the attention results

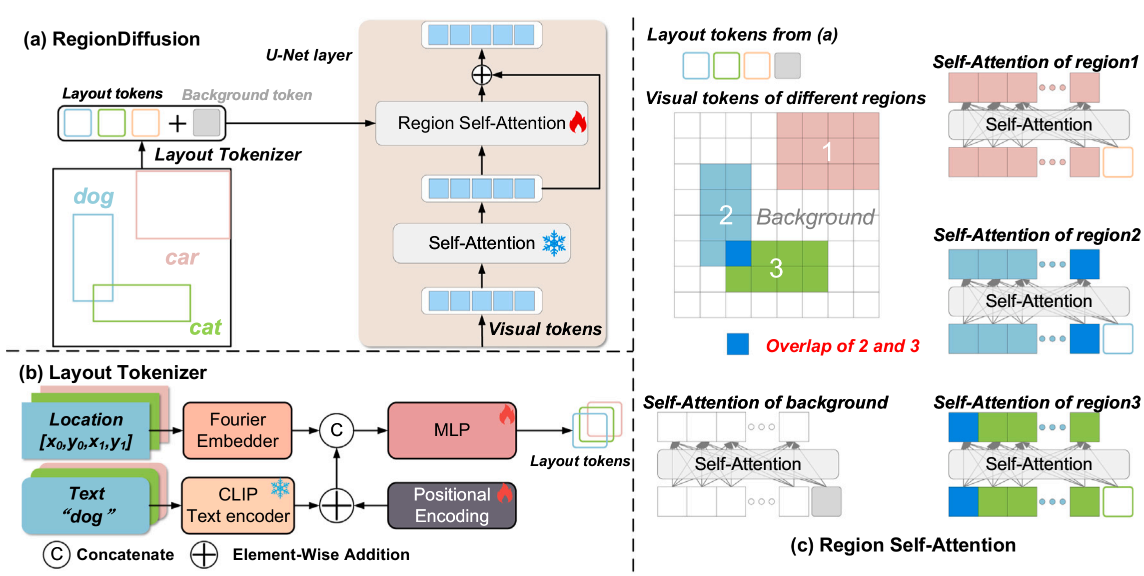

Fig. 1. Overview of the RegionDiffusion model: (a) RegionDiffusion integrates layout information into the U-Net layers of the diffusion model via Region Self-Attention; (b) Layout Tokenizer encodes the position and category information of each instance and fuses them into layout tokens; (c) Region Self-Attention divides visual features into multiple regions based on the layout and integrates the layout features into the visual features in the same region.

from all regions. The process of region self-attention is:

$$\mathbf{V}_i = \mathbf{V}_i + \tanh(\beta) \cdot \mathrm{Select}(\mathrm{SelfAttn}([\mathbf{V}_i, \mathbf{L}_i])) \tag{5}$$

$$\mathbf{V}_{bg} = \mathbf{V}_{bg} + \tanh(\beta) \cdot \mathrm{Select}(\mathrm{SelfAttn}([\mathbf{V}_{bg}, \mathbf{L}_{bg}])) \tag{6}$$

$$\mathbf{V} = \mathbf{V}_1 \cup \mathbf{V}_2 \cdots \cup \mathbf{V}_n \cup \mathbf{V}_{bg}, \tag{7}$$

where $\mathrm{Select}(\cdot)$ represents only select the visual tokens from the all tokens, β is a learnable parameter initialized at 0.

Regarding the overlapped token of multi-regions, as shown in Fig. 1(c), for the visual token $v_{overlap}$ in the overlapping part of Region 2 and Region 3: since $v_{overlap}$ belongs to both Region 2 and Region 3, it can communicate simultaneously with all other visual and layout tokens in Region 2 and Region 3. That is, it calculates the attention scores of $v_{overlap}$ with respect to these tokens and performs updates. For other tokens in Region 2 and Region 3, they can only communicate with tokens that belong to the same region.

3.3. Area-based loss re-weighting

The original simplified loss function of LDMs that calculates the error of each latent variable on average causes the model to focus more on large instances and background (which contribute more to the loss) during training. To prevent the model from overfocusing on the background and ignoring the generation of small instances, and to accelerate the training process, following the [11], we re-weight the loss based on the size of the instance regions' area. For the latent feature map $\mathbf{Z} \in \mathbb{R}^{H\times W\times C}$ obtained from the U-Net, we construct a re-weighting mask $\mathbf{M}' \in \mathbb{R}^{H\times W}$, where the value of each element $\mathbf{M}'_{i,j}$ in $\mathbf{M}'$ is defined as follows:

$$\mathbf{M}'_{i,j} = \begin{cases} (H\times W)^{-p}, & \text{if } (i,j) \in \text{Background} \\ w \times c_{i,j}^{-p}, & \text{if } (i,j) \in \text{Instances}, \end{cases} \tag{8}$$

where the w is set to 2, p is set to 0.2, and $c_{i,j}$ represents the area of the bounding box of the instance to which pixel (i,j) belongs. Finally, the mask $\mathbf{M}'$ is normalized by:

$$\mathbf{M}_{i,j} = H \times W \times \mathbf{M}'_{i,j} / \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{M}'_{i,j}, \tag{9}$$

and the final training loss of RegionDiffusion is:

$$\min_{\theta'} \mathcal{L} = \mathbb{E}_{\mathbf{z},\epsilon\sim\mathcal{N}(\mathbf{0},\mathbf{I}),t}\left[\|\epsilon - f_{\{\theta,\theta'\}}(\mathbf{z}_t, t, \mathbf{P}, \boldsymbol{L})\|_2^2\right] \odot \mathbf{M}, \tag{10}$$

where θ denotes the frozen parameters of the pre-trained LDM, θ' represents the newly introduced trainable parameters, $\mathbf{P}$ is the text prompt, $\boldsymbol{L}$ is the layout condition, and $\odot$ represents the Hadamard product.

4. Generative data augmentation pipeline

This section introduces a generative data augmentation pipeline specifically designed to generate images and corresponding annotations for training object detectors. The pipeline of generative data augmentation for object detection is illustrated in Fig. 2: (1) Remove bounding boxes with excessive overlap by Front Filter from annotations in the real dataset to reduce instance fusion during the generation; (2) Construct text prompts based on filtered annotations and captions; (3) Generate images based on the text prompts and annotations with RegionDiffusion. (4) Apply Post Filter to filter out instances that are not generated correctly from annotations; (5) Train the object detector using the generated images and filtered annotations.

4.1. Front filter based on NMS

High-quality image-annotation pairs are the cornerstone of object detection. Although we have adopted many methods in RegionDiffusion to make the generated instances match the given annotations, the model will still experience instance fusion and attribute leakage sometimes when given very complex layout conditions based on annotations from the real dataset, especially when there are a large number of overlapping bounding boxes. To reduce these issues, we use Non-Maximum Suppression (NMS) to filter out the bounding boxes with a high degree of overlap, and thus reduce the complexity of the overall layout conditions before giving the layout to RegionDiffusion. Algorithm 1 shows the front filtering process for the annotation containing N bounding box-category pairs, where the $\mathrm{IoU}(\cdot,\cdot)$ is the Intersection over Union between two bounding boxes.

4.2. Prompt construction

Text prompts are indispensable for layout-to-image diffusion models. Due to the maximum limit on token numbers of the CLIP text encoder, a simple and effective method is used to construct prompts: We directly connect the categories of instances together using "*and*" (discussions in Section 5.3). Meanwhile, in order to meet the generation requirements under different scene or artistic styles, we add the required scene to the end of the prompts, such as "*real world*", "*clip art style*", "*water-color style*", etc. In addition, for the dataset with captions, we incorporate them into the text prompts. In summary, we use the template "An image of <*category* 1> and <*category* 2> …and <*category* N> in <*scene*> scene where <*caption*>." to construct text prompts. For each instance, we use its category and box coordinates as layout conditions, then use text prompts and layout conditions to generate images through the RegionDiffusion mentioned in Section 3 to obtain the corresponding image-annotation pairs.

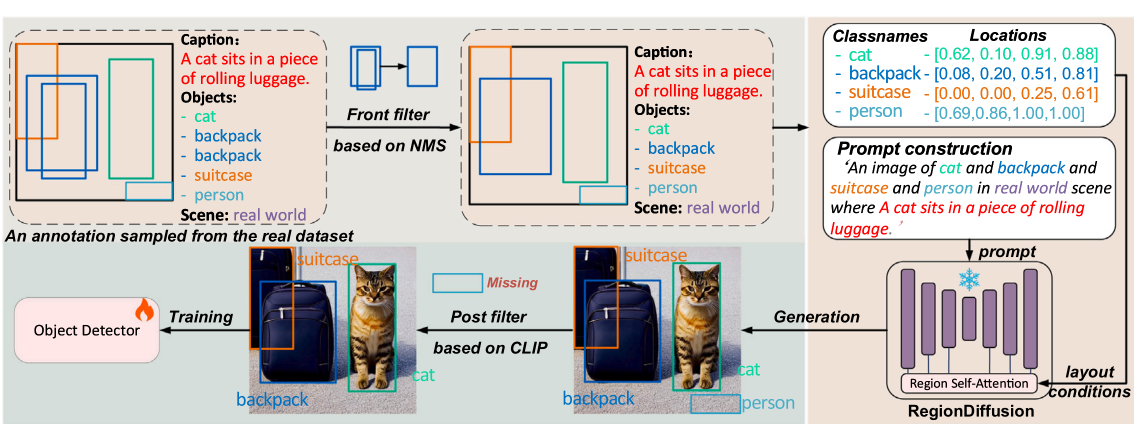

Fig. 2. The generative data augmentation pipeline. The entire pipeline consists of five steps: (1) applying a front filter based on NMS; (2) constructing prompts; (3) generating data using RegionDiffusion; (4) applying a post filter based on CLIP; and (5) training the detector with the generated data.

Algorithm 1 Annotation front filter based on NMS.

Input: Annotation $\mathcal{A} = \{(\mathbf{b}_1, c_1), ., (\mathbf{b}_N, c_N)\}$, NMS threshold $\mathcal{T}$, function $\mathrm{IoU}(\cdot,\cdot)$
Output: Filtered annotation $\mathcal{D}$
Initialize: Filtered Annotation $\mathcal{D} = \{\}$ ▷ Initialize empty set for selected boxes
while $\mathcal{A}$ not empty **do**
  $(\mathbf{b}_m, c_m) \leftarrow \mathcal{A}[0]$ ▷ Select next candidate box
  $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{b}_m, c_m)\}$ ▷ Add current box to results
  $\mathcal{A} \leftarrow \mathcal{A} - \{(\mathbf{b}_m, c_m)\}$ ▷ Remove selected box from candidates
  for $(\mathbf{b}_i, c_i)$ in $\mathcal{A}$ **do**
    if $\mathrm{IoU}(\mathbf{b}_m, \mathbf{b}_i) \geq \mathcal{T}$ **then**
      $\mathcal{A} \leftarrow \mathcal{A} - \{(\mathbf{b}_i, c_i)\}$ ▷ Suppress overlapping box
    end if
  end for
end while
Return $\mathcal{D}$ ▷ Return final filtered boxes

4.3. Post filter based on CLIP

Besides the aforementioned problems, the generated data occasionally has issues with the instance disappearance and instance category errors, so we propose a Post Filter based on CLIP scores to filter out incorrectly generated instances from annotations. For each of the N bounding box-category pairs $(\mathbf{b}_i, c_i)$, the filtering process is as follows: (1) Use the bounding box $\mathbf{b}_i$ to crop the corresponding instance from the generated image; (2) Compute the similarity score s_i between the cropped instance and its specified category c_i using CLIP; (3) Rank all pairs based on their scores s_i and remove the αN pairs with the lowest CLIP scores, where α is the filtering ratio between 0 and 1; Only the remaining $(1-\alpha)N$ bounding box-category pairs are retained as annotations for the generated image.

5. Experiments

We mainly conduct experiments in three aspects: **Fidelity**, **Zero-shot**, and **Trainability**. Fidelity measures the ability of layout-to-image generation. Zero-shot evaluates the generalization capability of the model, and further indicates whether it can generate object detection data across different domains without fine-tuning. Trainability examines the extent to which generated data by the model improves the performance of object detectors.

5.1. Experimental setup

5.1.1. Implementation details

We train RegionDiffusion based on the Stable Diffusion v1.5, which is pre-trained on the LAION-5B dataset [60]. To improve layout accuracy, we insert Region Self-Attention into the U-Net layers with resolutions of 64×64, 32×32, and 16×16, while keeping all the original parameters of Stable Diffusion frozen. We train RegionDiffusion for 80 epochs using the AdamW [61] optimizer with constant learning rate of 1e-4, and warm-up for the first 5k steps. The total batch size is 64, which requires 4 A100 GPUs with 40 GB VRAM. During the inference, we use the PLMS [62] scheduler to sample for 50 steps with setting classifier-free guidance [63] as to 7.5.

5.1.2. Training data

We only train our model on the COCO2017-train [16]. The text prompts are constructed using the template in Section 4.2 with the captions. During training, we apply random cropping to a size of 512×512 and random horizontal flipping with a 0.5 probability for image-annotation pairs. Unlike other methods, we do not exclude small objects, allowing our model to generate small instances well.

5.1.3. Evaluation data

For Fidelity, we use all images and annotations except for *is_crowd* instances from COCO2017-val. For Zero-shot, we validate using the challenging LVIS-minival [64] dataset, which contains 1,203 long-tail categories. For Trainability, in the general domain, we use COCO-10K (randomly sampling 10 k images from COCO2017-train to simulate the scale of typical custom training datasets) to train object detectors and evaluate the detectors using COCO2017-val. Additionally, we select five datasets with different domains to demonstrate the effectiveness of our augmentation method for downstream object detection tasks: VOC2007 [65], Clipart [14], Watercolor [14], Animals [66], and Excavators [66].

5.1.4. Evaluation metrics

For Fidelity, we use the Fréchet Inception Distance (FID) [67] to evaluate the quality of images and the Local-CLIP [68] to measure the consistency between the generated instance and category text. Regarding the accuracy of the positions of the generated instances, similar to [8,11], we use the YOLO scores. Specifically, we choose the YOLOv8m [15] pre-trained on COCO2017 to detect the objects in the generated images and report the APs, Recall and Precision. For Zero-shot, we use Co-DETR [69] pre-trained on LVIS to report the metrics on LIVS benchmark. For trainability, we select three mainstream detectors with different architectures: YOLOv8m, Faster R-CNN [30] and DINO [70] (with ResNet50 [71] pretrained on ImageNet1k [72] as backbone) to report the COCO's official metrics.

5.1.5. Baselines

We select several SOTA layout-to-image generation models for comparison in the three aspects mentioned above. **Training-free methods:** BoxDiff [41], R&B [42] and GrounDiT [43]; **Training-based methods:** ReCO [9], HiCo [10], MIGC [39], GLIGEN [8] and the method designed to generate data for object detection: GeoDiffusion [11]. **We only use our generative data augmentation pipeline in comparison of Trainability.**

5.2. Comparison with SOTAs

5.2.1. Fidelity

Quantitative results. We report the upper-bounds of each metric obtained by real images in Table 1. Our method surpasses all baselines on most metrics, especially on the YOLO scores, by a significant gap. Specifically, we compare the generation effects of small objects. The $\mathrm{AP}_{\mathrm{small}}$ shows that most methods can't properly generate small objects and thus can't enhance detectors' ability to detect small targets well.

Qualitative results. As shown in the qualitative analysis in Fig. 3, in the first row, under complex conditions with numerous small and overlapping objects, our method generates every object perfectly, while others are unable to handle such complex generation scenarios with issues like messy results, objects missing, and wrong or even failed generation. After setting layout conditions, models may generate related objects not present in the original layout, which we define as the "illusions". These "illusions" introduce extra objects that are not in the annotation, affecting detector performance as they are wrongly classified as background. In the second row, we generate a clear background but other methods have "illusions" to varying degrees in generating extra objects. The visual quality and layout conformity are crucial for layout-to-image generation models. In the third row, our method perfectly generates even the most challenging "person" object realistically, while others produce distortions. More qualitative results are shown in the Appendix D.

5.2.2. Zero-shot

To evaluate the zero-shot ability of all methods, we conduct experiments on the LIVS-minival dataset. Table 2 shows that although our model is only trained on the COCO dataset, due to the fact that all the original parameters of pre-trained Stable Diffusion [19] are frozen

Table 1
Quantitative results compared with SOTAs in terms of the fidelity on COCO2017-val. RegionDiffusion achieved the best performance across all metrics, except for Local CLIP. *: Uses the last token of every classname as the guide, since it can only use one token as guidance for each instance. †: Combined with GLIGEN.

Method	Venue	AP ↑	AP_{50} ↑	AP_{75} ↑	AP_{small} ↑	ReCall ↑	Precision ↑	FID ↓	Local CLIP ↑
Real Image	–	50.1	66.7	54.6	32.3	61.0	71.6	–	19.8
Training-free									
R&B [42]	ICLR2024	3.3	9.8	1.6	0.1	14.6	26.3	31.5	17.9
BoxDiff* [41]	ICCV2023	1.7	5.2	0.8	0.0	8.5	17.3	31.8	17.8
BoxDiff† [41]	ICCV2023	18.3	33.7	17.7	1.0	34.3	50.9	29.8	18.6
GrounDiT [43]	NeurIPS2024	4.3	12.8	1.8	0.1	17.9	28.2	36.1	18.3
Training-based									
ReCo [9]	CVPR2023	31.2	58.3	30.0	11.5	56.0	65.0	29.6	**19.5**
HiCo [10]	NeurIPS2024	13.6	22.6	12.6	1.8	29.5	38.4	31.2	18.3
MIGC [39]	CVPR2024	22.1	41.8	21.2	1.8	40.9	60.4	33.9	18.7
GLIGEN [8]	CVPR2023	19.3	34.8	19.2	2.0	36.0	52.3	28.7	18.7
GeoDiffusion [11]	ICLR2024	28.7	42.0	31.2	1.5	39.2	62.5	27.6	18.6
Ours	This paper	**37.3**	**60.4**	**37.9**	**18.5**	**57.3**	**71.1**	**27.1**	19.0

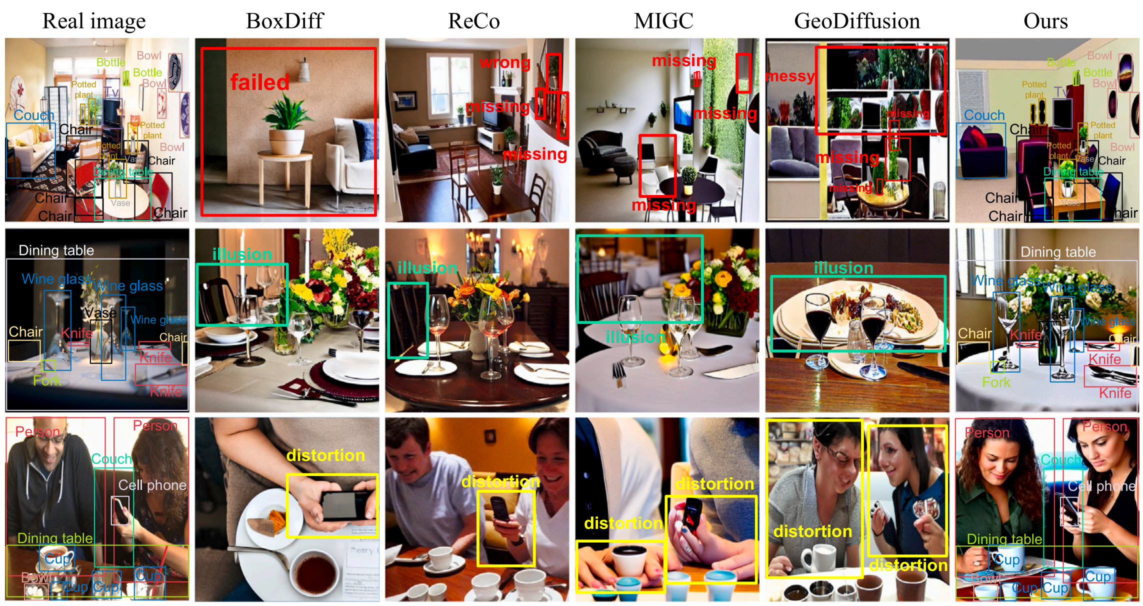

Fig. 3. Quantitative results on COCO2017-val. 1st row: Bold red bounding boxes mark incorrectly-generated instances (e.g., failed, missing, messy and wrong); 2nd row: Bold green boxes indicate extra-generated instances (illusions); 3rd row: Bold yellow boxes show distorted instances. Real and our images are annotated to show that our method has extremely high layout accuracy and visual quality.

Table 2
Results of Zero-shot on LVIS-minival. We use the annotations of the LVIS-minival as conditions to generate images and then employ Co-DETR (with Swin-L [73] as the backbone) pre-trained on LVIS to report the metrics. *: The images were resized to 512 × 512 before evaluation. †: GLIGEN is trained on GoldG [74], O365 [75], SBU [76] and CC3M [77], while other training-based methods are only trained on COCO.

Method	AP	AP_{50}	AP_{75}	AP_s	AP_r	AP_c	AP_f
Real Image*	59.8	73.2	63.5	48.2	48.6	60.9	60.9
Training-free							
R&B [42]	1.3	3.9	0.7	0.1	1.9	1.5	1.0
BoxDiff [41]	1.5	3.8	1.2	0.0	1.4	2.1	1.1
GrounDiT [43]	3.1	8.7	1.6	0.1	1.3	4.6	2.2
Training-based							
ReCo [9]	13.1	**25.8**	12.0	5.8	7.5	15.0	12.4
HiCo[10]	9.0	15.3	8.7	1.6	6.0	11.9	7.0
MIGC [39]	12.5	24.2	11.3	1.4	9.0	14.3	11.6
GLIGEN† [8]	10.5	19.5	10.2	1.1	9.2	11.7	9.8
GeoDiffusion [11]	4.7	7.1	5.1	0.3	2.6	3.1	6.5
Ours	**14.7**	23.1	**15.3**	**8.0**	**14.0**	**15.9**	**13.7**

during training, our model still has the outstanding zero-shot ability to generate objects in the open world. Furthermore, our method can be applied to datasets of different domains without any fine-tuning. However, GeoDiffusion has the worst zero-shot ability except for the training-free method BoxDiff because it fine-tunes the text encoder and U-Net, resulting in severe knowledge forgetting.

5.2.3. Trainability

On the COCO-10K, we first evaluate the results of YOLOv8 and DINO trained using only real data as the baseline performance without augmentation. To compare the enhancement effects of generated data from different models on object detection tasks, we use baselines to generate additional 1× data with the annotations of real dataset and train object detectors using both the real and generated data. As shown in Tables 3 and 4, even when training the detectors using only original annotations and images generated by RegionDiffusion without the pipeline (Ours), our method still achieves the best results on all three detectors. When it comes to the combination of RegionDiffusion and Pipeline (Ours + Pipeline), it achieves the highest enhancement effects on both YOLOv8

Table 3
Trainability results of all methods for YOLOv8m on COCO-10K. YOLOv8m is trained from scratch for 100 epochs following the default settings. †: Random cropping and horizontal flipping (probability 0.5) are applied to enhance layout diversity during generation.

Method	Scale	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
YOLOv8m		25.5	38.1	27.3	12.1	26.8	34.0
w/ BoxDiff	1×	26.0 (0.5↑)	38.3	27.6	12.4	27.2	35.5
w/ Reco	1×	27.9 (2.4↑)	41.5	29.8	14.4	29.9	37.2
w/ MIGC	1×	28.1 (2.6↑)	41.3	30.2	13.2	29.7	39.1
w/ GLIGEN	1×	27.8 (2.3↑)	41.1	29.7	13.5	29.0	39.1
w/ GeoDiff	1×	28.6 (3.1↑)	41.9	30.5	12.8	30.3	40.2
w/ Ours	1×	**29.1** (3.6↑)	**42.3**	**30.8**	**13.4**	**30.5**	**41.4**
w/ Ours + Pipeline	1×	**29.3** (3.8↑)	**42.9**	**31.3**	**14.0**	**30.8**	**41.7**
w/ Ours + Pipeline†	2×	**30.7** (5.2↑)	**44.9**	**32.7**	**14.5**	**32.5**	**43.3**
w/ Ours + Pipeline†	3×	**31.6** (6.1↑)	**45.8**	**33.7**	**15.3**	**33.8**	**44.1**

Table 4
Trainability results of all methods for Faster R-CNN and DINO detectors on COCO-10K dataset. Faster R-CNN and DINO are trained from scratch for 12 epochs (schedule 1×) following the default settings.

Method	Scale	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
Faster-RCNN		21.9	41.2	20.9	11.6	24.2	28.4
w/ BoxDiff	1×	21.2 (0.7↓)	40.2	20.2	11.5	23.6	27.4
w/ ReCo	1×	23.0 (1.1↑)	42.4	22.6	11.7	25.5	30.7
w/ MIGC	1×	23.0 (1.1↑)	41.8	22.9	11.5	25.0	31.5
w/ GLIGEN	1×	23.2 (1.3↑)	42.4	22.9	11.6	25.3	31.5
w/ GeoDiff	1×	24.2 (2.3↑)	43.4	24.1	11.9	26.7	**32.9**
w/ **Ours**	1×	**24.5** (2.6↑)	**43.6**	**24.6**	**12.5**	**26.8**	32.6
DINO-R50		26.3	39.5	27.9	14.4	28.7	35.5
w/ BoxDiff	1×	25.5 (0.8↓)	38.2	27.1	14.0	27.7	34.2
w/ ReCo	1×	29.3 (3.0↑)	44.5	31.1	16.0	31.8	39.1
w/ MIGC	1×	29.4 (3.1↑)	44.3	31.4	15.1	31.5	39.6
w/ GLIGEN	1×	29.6 (3.3↑)	44.1	31.3	15.0	31.6	40.6
w/ GeoDiff	1×	31.0 (4.7↑)	45.9	32.9	16.1	33.2	**42.7**
w/ **Ours**	1×	**31.3** (5.0↑)	**46.5**	**33.1**	**16.2**	**33.6**	41.0
w/ **Ours + Pipeline**	1×	**31.6** (5.3↑)	**47.0**	**33.5**	**16.4**	**34.3**	41.9
w/ GeoDiff	2×	32.6 (6.3↑)	47.7	34.6	16.0	35.2	44.5
w/ Ours	2×	33.1 (6.8↑)	49.1	35.2	18.5	35.3	44.7

Table 5
Trainability on 5 downstream datasets of YOLOv8m trained from scratch and pre-trained on COCO2017. We also report the differences in APs before and after using generated data.

		From scratch			Pre-trained		
Dataset	Method	AP	AP_{50}	AP_{75}	AP	AP_{50}	AP_{75}
VOC2007	YOLOv8m	43.1	63.5	46.2	58.6	78.0	63.5
	w/ Ours	**51.5**	**72.4**	**55.4**	**59.8**	**80.1**	**64.4**
		8.4↑	8.9↑	9.2↑	1.2↑	2.1↑	0.9↑
Clipart	YOLOv8m	12.1	21.7	11.7	32.9	51.8	35.4
	w/ Ours	**24.8**	**40.9**	**27.3**	**36.5**	**58.9**	**38.8**
		12.7↑	19.2↑	15.6↑	3.6↑	7.1↑	3.4↑
Watercolor	YOLOv8m	11.9	22.8	10.8	20.3	37.8	20.6
	w/ Ours	**21.6**	**39.9**	**19.9**	**30.8**	**53.7**	**31.0**
		9.7↑	17.1↑	9.1↑	10.5↑	15.9↑	10.4↑
Animals	YOLOv8m	44.6	63.6	48.1	66.8	86.5	74.5
	w/ Ours	**62.8**	**82.8**	**68.6**	**70.0**	**89.5**	**77.3**
		18.2↑	19.2↑	20.5↑	3.2↑	3.0↑	2.8↑
Excavators	YOLOv8m	78.0	91.7	88.3	79.7	93.1	88.0
	w/ Ours	**78.6**	**92.4**	**88.6**	**80.8**	**93.4**	**90.2**
		0.6↑	0.7↑	0.3↑	1.1↑	0.3↑	2.2↑

Table 6
The impacts of region self-attention and area re-weighting loss. *: replace the region self-attention with the vanilla self-attention. RSA-64 denotes applying region self-attention on the layers with a resolution of 64×64.

RSA-64	RSA-32	RSA-16	Area loss	AP ↑	AP_{50} ↑	AP_{75} ↑	Recall ↑	FID ↓
*	*	*		0.8	2.6	0.4	5.0	30.7
✔				17.6	32.0	16.8	34.8	27.6
✔	✔			20.0	35.2	19.1	36.4	**26.1**
✔	✔	✔		22.5	39.7	21.8	42.0	**26.1**
✔	✔	✔	✔	**24.6**	**42.0**	**24.2**	**42.9**	26.3

Table 7
Impact of different prompt construction strategies on generation fidelity.

Real captions	Pseudo captions	AP ↑	AP_{50} ↑	AP_{75} ↑	Recall ↑	FID ↓
✔	✗	25.7	45.0	25.2	42.4	**27.8**
✗	✔	**27.2**	47.9	26.1	43.1	29.5
✔	✔	**27.2**	**48.1**	**26.5**	**43.5**	28.8

and DINO (+ 3.8 AP and + 5.3 AP respectively), demonstrating the superiority of our approach. Furthermore, we increase the scale of generated data to 2× and 3×. As the scale of data augmentation increases, our method brings greater improvements to YOLOv8's performance (+ 5.2 AP for 2× and + 6.1 AP for 3×). However, we observe that our method underperforms GeoDiffusion in terms of AP_l. Therefore, we increase the scale of the synthetic data to 2×. As shown in the last two rows of Table 4, we successfully achieve the state-of-the-art performance across all metrics.

Table 5 demonstrates the effectiveness of our method for downstream tasks without the need for further fine-tuning. Whether training the detector from scratch or pre-trained on COCO2017, our method improves YOLOv8's performance across datasets of different scales, artistic styles, and categories. The experiments show that our method significantly improves the performance of object detectors in both general and specific domains.

5.3. Ablation study

5.3.1. Region self-attention and loss re-weighting

As shown in Table 6, the first row uses vanilla self-attention instead of Region Self-Attention as the fusion mechanism, failing to properly learn layout information within the same training time. By deploying Region Self-Attention in more different resolution layers of the U-Net, the model achieves better positional accuracy. After applying Area-based Loss Re-weighting, although the FID slightly deteriorates, the positional accuracy improves significantly.

5.3.2. Prompt construction strategies

To investigate the impact of text prompt construction methods on model generation, we train and evaluate the model with three strategies: (1) only real captions from COCO, (2) only pseudo-captions generated with the template "An image of <*category* 1> and <*category* 2> …and <*category* N>.", and (3) combining both. As shown in Table 7, pseudo-captions describe all objects and thus achieve higher YOLO scores, while real captions focus on fine-grained attribute descriptions, leading to better quality. Combining both approaches balances the positional accuracy and quality effectively.

5.3.3. Front filter and post filter in the pipeline

We apply the Front Filter with different NMS thresholds for data augmentation on COCO2017-val. As shown in Table 8, using augmented data improves performance even without the Front Filter (threshold of 100 %). The best results occur at a threshold of 0.4, while lower thresholds overly filter objects, reducing performance. Then, we apply the Post Filter with different α settings mentioned in Section 4.3. As shown in Table 9, our Post Filter achieves the best performance after filtering out 5 % of the bounding box-category pairs with lowest CLIP scores.

Table 8
Impact of different NMS thresholds of the front filter on the trainability. Real only represents that the detector is trained only on real data.

IoU threshold	Real only	100 %	70 %	60 %	50 %	40 %	30 %
AP	8.7	18.3	18.3	18.3	18.5	**18.7**	18.5
AP_{50}	14.9	29.2	29.3	29.0	29.4	**29.5**	29.3
AP_{75}	8.8	19.0	18.9	18.9	19.3	**19.4**	19.3

Table 9
Impact of different filtering ratios α of the post filter on the trainability.

α	Real only	0 %	5 %	10 %	20 %	30 %
AP	8.7	18.7	**18.8**	18.7	18.5	18.1
AP_{50}	14.9	29.5	**29.6**	29.5	29.3	29.0
AP_{75}	8.8	**19.4**	**19.4**	19.3	19.3	18.9

Table 10
The performance of YOLOv8m on the COCO10k dataset under joint training with different ratios of real data and generated data.

Training data	AP	AP_{50}	AP_{75}	AP_s	AP_m	AP_l
100 % real + 0 % generated	25.5	38.1	27.3	12.1	26.8	34.0
0 % real + 100 % generated	9.5	15.4	9.6	2.7	8.9	16.4
25 % real + 100 % generated	20.9	32.3	21.9	8.8	21.4	30.6
50 % real + 100 % generated	24.4	36.9	25.9	11.7	25.3	34.8
75 % real + 100 % generated	27.0	40.2	28.7	12.5	28.2	38.0
100 % real + 100 % generated	**29.3**	**42.9**	**31.3**	**14.0**	**30.8**	**41.7**

5.4. Real data is indispensable for training object detectors

Although our method can generate high-quality data for object detection tasks, in the experiments, we found that manually and meticulously annotated data is still unsurpassable and irreplaceable. We specifically used our pipeline to generate synthetic data for the COCO-10K dataset, with the same quantity as the real data. Then, we jointly used different ratios of real data with all the generated data to train YOLOv8m from scratch and evaluated its performance. As shown in Table 10, the results obtained by using only the generated data are far inferior to those achieved by using 100 % real data. When the ratio of real data increases, the performance of the detector also improves, and there is no sign of saturation. This indicates that generated data alone cannot be used to train object detectors, and real data is indispensable for training object detectors.

Table 11
The impact of inserting different types of self-attention and the number of generated objects on the computational overhead of the model.

Model	Objects	Params (B)	FLOPs (T)	Inference time (s)	VRAM (G)
Stable Diffusion v1.5	–	1.07	20.10	6.59	9.51
+Vanilla Self-attention	1	1.28	28.29	11.02	10.30
+Region Self-attention	1	1.28	28.29	12.31	10.98
+Region Self-attention	5	1.28	28.32	12.58	11.15
+Region Self-attention	10	1.28	28.35	12.74	12.31
+Region Self-attention	20	1.28	28.40	13.13	12.42

5.5. The impact of RegionDiffusion on computational overhead

Inserting attention layers into the original Stable Diffusion model increases model parameters and requires layout information preprocessing, resulting in extra computational latency and memory usage. We analyze the impact of different attention layers and object counts on parameters, FLOPs, inference time, and VRAM usage. For fairness, all tests use FP32 precision without inference optimizations.

Table 11 shows that compared with SDv1.5, RegionDiffusion adds small amount of parameters and memory but incurs some extra FLOPs and inference time. Compared with Vanilla Self-Attention, Region Self-Attention computes multiple regions in parallel, with minimal extra cost beyond region division/integration. More objects increase layout tokens in self-attention, causing slight latency growth.

5.6. Application in cross-domain object detection

We also extend our method to the cross-domain object detection task to demonstrate that our method can be applied to various downstream tasks. We take the cross-domain object detector H^2FA R-CNN [78] (with backbone of ResNet-101) as the target of our generative data augmentation method, and conducted experiments with the VOC dataset as the source domain and the Clipart and Watercolor datasets as the target domains. For the two target domains, we used RegionDiffusion and Pipeline to generate an additional set of images with double the quantity as the auxiliary training set. To evaluate the effect, we reported the

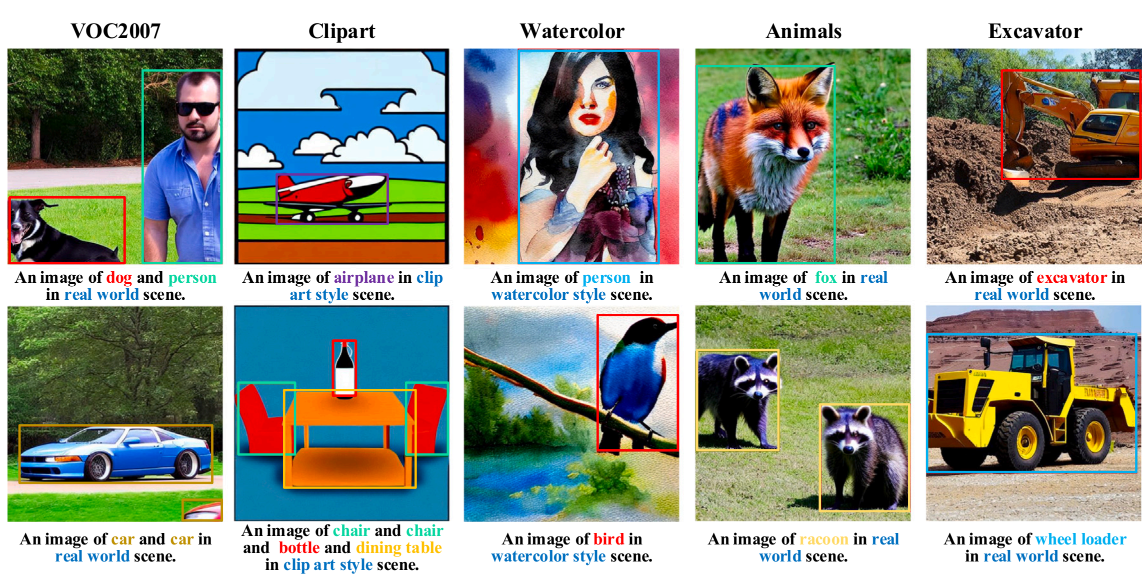

Fig. 4. Generated images for five downstream object detection datasets by RegionDiffusion without fine-tune.

Table 12
Experimental results of our data augmentation method applied to cross-domain object detection tasks. †: Results obtained by our re-implementation on Detectron2 v0.6 [79].

Dataset	Method	AP	AP_{50}	AP_{75}	CL	CL_{50}	CL_{75}
$Clipart_{test}$	source only		29.2				
$Clipart_{test}$	H^2FA R-CNN†	26.79	52.51	23.60	37.76	70.59	36.05
$Clipart_{test}$	H^2FA R-CNN + Ours	**27.80**	**53.77**	**25.96**	**39.78**	**71.88**	**40.71**
Watercolor	source only		41.1				
Watercolor	H^2FA R-CNN†	29.48	57.58	**26.51**	43.04	75.79	42.91
Watercolor	H^2FA R-CNN + Ours	**29.82**	**58.18**	24.92	**43.55**	**76.80**	**43.60**

Table 13
Performance comparison of YOLO-World w/o our data augmentation on open-vocabulary tasks.

Dataset	Model	AP	AP_{50}	AP_{75}	Precision	Recall
Animals	YOLOv8m	44.6	63.6	48.1	66.6	55.0
Animals	YOLOv8m + Ours	62.8	82.8	68.6	84.9	73.1
Animals	YOLO-World-M (Zero-shot)	44.7	56.8	49.5	64.9	51.7
Animals	YOLO-World-M (Fine-tune)	70.5	88.7	76.7	86.5	82.7
Animals	**YOLO-World-M + Ours**	**72.5**	**90.7**	**78.5**	**89.7**	**83.5**
Excavators	YOLOv8m	78.0	91.7	88.3	91.2	83.0
Excavators	YOLOv8m + Ours	78.6	92.4	88.6	92.8	84.1
Excavators	YOLO-World-M (Zero-shot)	40.0	51.6	45.8	40.9	73.1
Excavators	YOLO-World-M (Fine-tune)	81.1	92.2	88.5	**93.0**	**87.1**
Excavators	**YOLO-World-M + Ours**	**81.7**	**93.8**	**90.3**	92.9	87.0

commonly used bbox APs and CorLocs in cross-domain object detection tasks.

As shown in Table 12, the performance of H^2FA R-CNN has been significantly improved after applying our proposed generative data augmentation method, which indicates that our method can also be used to assist in cross-domain object detection tasks.

5.7. Discussion on open-vocabulary object detection task

In Section 5.2.2, we demonstrate that our method can generate objects beyond the 80 categories seen during training. We aim to further explore two aspects: compared with open-vocabulary object detectors, whether our method can generate new samples to help detectors achieve higher accuracy on more specific categories, and whether our method can be used to enhance the performance of open-vocabulary object detectors. We select the YOLO-World [80], an open-vocabulary object detector based on YOLOv8, and conduct experiments on the Animals and Excavators datasets, whose categories differ from those of COCO.

It can be observed in Table 13 that in the zero-shot setting, YOLO-World performs worse than YOLOv8 trained with our data augmentation method. However, when YOLO-World is fine-tuned on the corresponding datasets (fine-tuned) and then our method is incorporated, its performance further improves.

6. Conclusion

We propose RegionDiffusion, a novel layout-to-image model, and a generative data augmentation pipeline for object detection. By integrating Region Self-Attention and Area-based Loss Re-weighting, RegionDiffusion improves the visual quality and positional accuracy of generated data. Experiments across multiple datasets demonstrate its superiority over state-of-the-art models in fidelity, zero-shot, and trainability performance. The pipeline enhances object detector performance via front and post filters and can generate data for general and specific domains. We hope this research can enable the development of more sophisticated and versatile data generation frameworks.

CRediT authorship contribution statement

Kesong Li: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shang Luo:** Writing – original draft. **Kuo-Kun Tseng:** Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Limitation

Although our method exhibits certain zero-shot capabilities, constrained by the knowledge of Stable Diffusion, it is unable to directly perform data augmentation on some datasets of uncommon objects, such as those of industrial part defects, infected leaves, and brain CT scans. However, users can fine-tune RegionDiffusion on these datasets to enable it to generate the desired objects. In addition, the generative data augmentation method of layout-to-image has a positive promotional effect when the training data is relatively sparse and the detector has not achieved a high APs. Nevertheless, when the scale of the training data is large and already highly diverse (e.g., the complete COCO2017 dataset), the effect of data augmentation is relatively limited. Moreover, when the detector can already achieve extremely high performance on real data, as in the case of the Excavators dataset in Table 5, the improvement brought by our method is relatively small.

Appendix B. Details of experiments on downstream datasets

To test the enhancement effect of our method on downstream task datasets, we generated data for each dataset without fine-tuning. The details are shown in Table B.14. For the VOC2007, which has more than 5000 training images, we only generated 2× the amount of synthetic data. For other small-scale datasets, we generated 5× the amount of synthetic data and trained them together with the real data. Regarding the division of the training set and the test set, we followed the common division methods of each dataset. Except for the Animals and Excavators datasets, for which we combined the validation set and the test set because the number of images in their test sets was too small. To enable the model to generate data for Clipart and Watercolor, we set the scene to the same artistic style. Fig. 4 shows some examples of the data generated by our method for five downstream task datasets and the corresponding text prompts.

Appendix C. Differences between our method and MIGC

First, in terms of attention structure, we only use a single-layer self-attention mechanism with shared weights to handle the layout generation control between multiple instances and the background in one go, which we consider concise and efficient. However, MIGC performs various different attention calculations for the background and instances. For example, it requires Cross Attention and Layout Attention calculations for the background, while using Cross Attention and Enhancement Attention for instances.

Table B.14
Setup of augmentation pipeline for 5 downstream datasets.

Dataset	Scale	Train data	Test data	Scene
VOC2007	2×	train+val	test	*real world*
Clipart	5×	train	test	*clip art style*
Watercolor	5×	train	test	*watercolor style*
Animals	5×	train	test+val	*real world*
Excavators	5×	train	test+val	*real world*

Fig. A.5. More quantitative results on COCO2017-val.

Second, in terms of design philosophy, our idea is to divide the image into multiple sub-regions for generation, whereas MIGC divides the task from layout to image into multiple subtasks.

Finally, regarding the insertion method of the layout control module: since MIGC needs to perform three different attention calculations for each newly inserted module, this brings significant computational overhead. To reduce the computational load, MIGC only inserts controllers into the U-Net blocks with the lowest resolution (8×8). This results in insufficiently precise layout control because dividing regions at a low resolution leads to large errors compared to dividing regions on the original image, and small objects may even be ignored. In contrast, we choose to divide regions at larger resolutions (64×64, 32×32, 16×16) while discarding smaller resolutions to avoid inaccuracies in layout control.

Appendix D. More quantitative results

Fig. A.5 shows more results of qualitative comparison with baselines. Compared with other methods, the images generated by us have the highest visual quality, location accuracy, and attribute accuracy.

Data availability

Data will be made available upon request.

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